

Pipe Thickness Calculation Report

Minimum wall thickness due to internal pressure

Report type

**Engineering Calculation
Report**

Code basis

ASME B31.3 Process Piping

Generated

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1 Calculation Context

Objective	Verify minimum required nominal thickness for HP steam pipe line as per ASME B31.3.
Project No. / MOC No. / Job No.	NRL/CDU-VDU/RTA-2027
Line No. / Tag No.	18"-SM-20-2301-B2A-IH

2 Executive Result Summary

Pressure thickness, t

11.90 mmThickness after corrosion
allowance, tm**13.40 mm**Required nominal
thickness**15.32 mm**

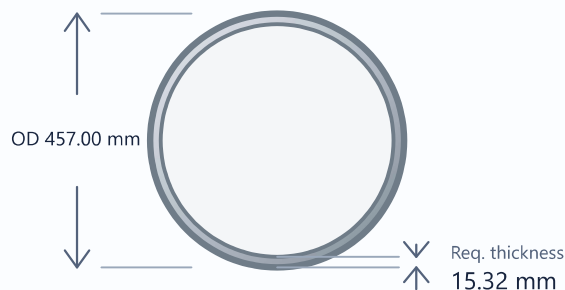
Suggested schedule

SCH 60

19.05 mm

2D Section View

SCH 60 · 19.05 mm



Final schedule library used	ASME B36.10M steel
Schedule thickness margin	3.73 mm
Schedule thickness margin	24.39%
Engineering decision status	ACTION REQUIRED

3 Design Input Basis

Design pressure	49 kg/cm ²
Converted pressure, P	4.8053 MPa
Design temperature	480 deg C
Nominal pipe size	18 inch
Outside diameter, D	457.00 mm
Material	ASTM A335 GR.P11
Pipe type	seamless
Schedule library	ASME B36.10M steel
Corrosion / mechanical allowance, c	1.50 mm
Mill tolerance	12.50%

4 Code Factors Used

Factor name	Symbol	Value	Verification requirement
Allowable stress	S	90.340 MPa	Verify from controlled ASME B31.3 stress table and project PMS.
Quality factor	E	1.000	Verify pipe product form, weld quality and applicable code table.
Weld joint strength reduction factor	W	1.000	Verify weld type, temperature and applicable B31.3 W-factor basis.
Y coefficient	Y	0.400	Verify material category and design temperature basis.

5 Equation and Calculation Flow

ASME B31.3 Eq. (304.1.1)

$$t = \frac{P \times D}{2 \times (S \times E \times W + P \times Y)} + C$$

P	Design pressure
D	Outside diameter
S	Allowable stress
E	Quality factor
W	Weld joint strength reduction factor
Y	ASME Y coefficient
C	Corrosion allowance

Step	Description	Formula / Basis	Thickness (mm)
1	Pressure thickness (t)	ASME Eq. 304.1.1	11.90
2	Add corrosion allowance (C)	$t_m = t + C$	13.40
3	Apply mill tolerance (12.50%)	$t_n = t_m / (1 - \text{tolerance})$	15.32
4	Nearest higher schedule	SCH 60	19.05



D/6 thin-wall limit	76.167 mm
$P/(S \times E)$	0.05319
Calculated pressure thickness, t	11.901 mm
Thickness after corrosion allowance, t_m	13.401 mm
Required nominal thickness	15.315 mm

6 Schedule Selection Summary

Primary schedule library checked	ASME B36.10M steel
Final schedule library used	ASME B36.10M steel
Suggested schedule	SCH 60
Selected schedule thickness	19.05 mm
Margin over required nominal thickness	3.73 mm (24.39%)

7 Engineering Checks and Actions

Category	Assessment	Status	Remarks / Action
Pressure Thickness	Thin-wall check	● PASS	Thin-wall check passed: calculated pressure thickness t is less than D/6.
Temperature Limit	ASTM material specification limit	● PASS	Material specification temperature limit check passed: ASTM A335 GR.P11 design temperature 480 deg C is within -30 to 482 deg C from astm_piping_material_specification_webapp row MS-019.

Category	Assessment	Status	Remarks / Action
Pressure Thickness	Pressure ratio check	● PASS	Pressure ratio check passed: $P/(S \times E)$ is within 0.385.
Pressure Thickness	Nearest higher standard thickness	● PASS	Nearest higher standard thickness found: SCH 60 at 19.05 mm from ASME B36.10M steel.
Temperature Limit	API 574 Table 7 check	● WARNING	API 574:2016 Table 7: design temperature 480 deg C is at/above the Table 7 limit of 205 deg C. Consult with your Engineering Department.
Stress Screening	Upfront stress screening result	● LEVEL 3	Level 3: Detailed CAESAR II stress analysis required. Do not finalize pipe schedule/routing without Piping Stress Engineer review. Matched basis: IBR / steam / BFW line above 2 inch; M - All other types except noted above: Level 3.
Stress Screening	Stress screening scope	● WARNING	Pipe thickness due to internal pressure is outside this stress-analysis specification. Calculate minimum required pipe thickness separately as per applicable piping code, then use this screening to identify stress-analysis, support, flexibility, nozzle-load, dynamic, wind, seismic, flange-leakage, and consultation requirements.
Code Basis	Inferred service	● WARNING	inferred service 'steam'. Code basis to verify: IBR / ASME B31.1 for Steam and BFW piping.
Flexibility / Stress	Bourdon effect screening	● WARNING	Bourdon effect warning: design pressure 48.05 bar is above 20 bar. This is a stress/flexibility modeling warning, not a pressure-thickness formula.
Seismic	Seismic analysis screening	● WARNING	Seismic analysis screening: project rule flags seismic analysis for lines 12 inch and above.
Flange Leakage	Flange leakage screening	● WARNING	Flange leakage screening: project rule flags flange leak check for 18 inch and above lines except air and water lines.
Seismic	Seismic analysis screening	● WARNING	Minimum pressure thickness alone is not final pipe acceptability. Pipe routing, supports, flexibility, nozzle loads, dynamic loads, wind, seismic, flange leakage, corrosion, testing, and vendor limits may govern final pipe specification or support design.
Input Governance	Auto-filled values check	● WARNING	Auto-filled values reduce typing effort, but final S, E, W and Y shall still be verified from the controlled ASME B31.3 edition, project PMS and approved design basis.
Calculation Scope	V1 calculation limitation	● WARNING	IBR applicability, B31.1 versus B31.3 basis, flexibility, branch reinforcement, bends, NDE, PWHT and hydrotest are outside this V1 calculation.

8 Engineering Notes

IBR HP steam service. Verify B31.3/B31.1/IBR governing basis, approved stress value, PMS, PWHT, NDE, flexibility, branch reinforcement and hydrotest separately.

9 Scope Limitation and Disclaimer

This tool is for preliminary engineering calculation support only. Final design shall be verified against the controlled code edition, project PMS, IBR requirements where applicable, and approved engineering judgment. This calculation does not replace flexibility analysis, branch reinforcement check, bend or fitting assessment, flange leakage check, NDE, PWHT, hydrotest, or formal design approval.

10 Verification and Approval

Role	Name	Signature	Date
Prepared by			
Checked by			
Approved by			